

Introduction

Exercises ISLR – Ch.12

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Conceptual

Exercise: 3

Ex.3) K-mean clustering $K=2$ (manually)

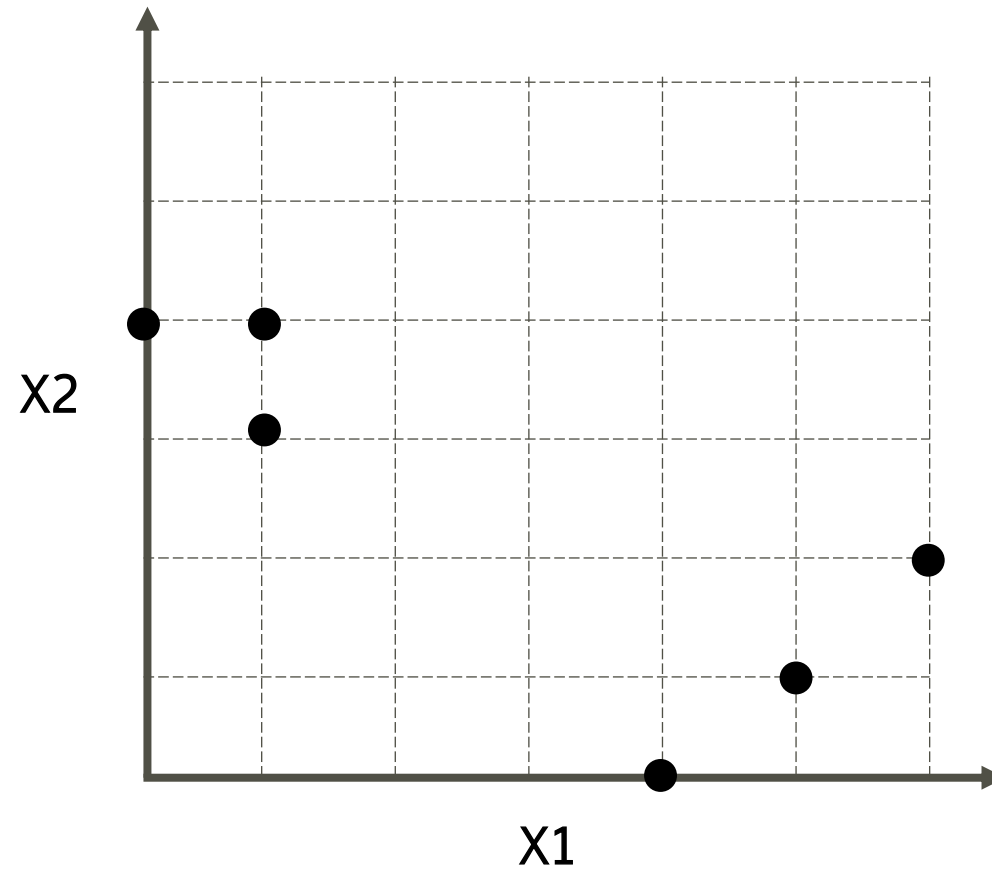
3. In this problem, you will perform K -means clustering manually, with $K = 2$, on a small example with $n = 6$ observations and $p = 2$ features. The observations are as follows.

Obs.	X_1	X_2
1	1	4
2	1	3
3	0	4
4	5	1
5	6	2
6	4	0

Ex.3) K-mean clustering K=2 (a)

(a) Plot the observations.

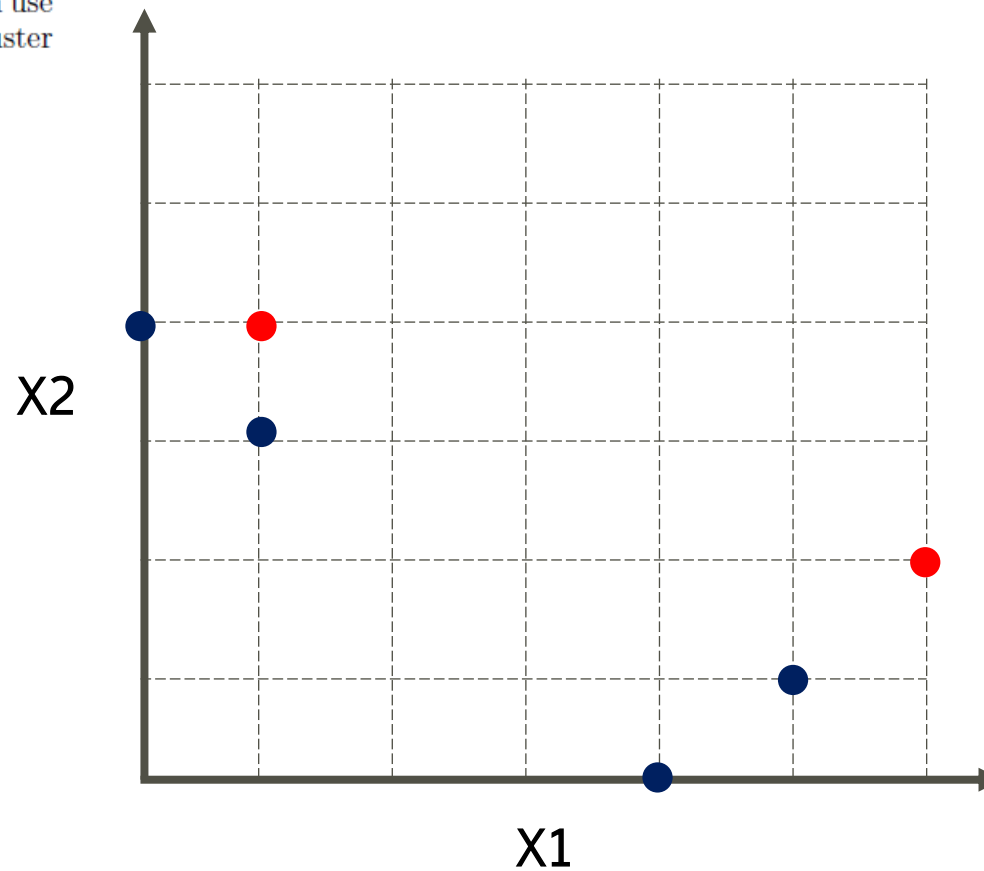
Obs.	X_1	X_2
1	1	4
2	1	3
3	0	4
4	5	1
5	6	2
6	4	0



Ex.3) K-mean clustering K=2 (b)

- (b) Randomly assign a cluster label to each observation. You can use the `np.random.choice()` function to do this. Report the cluster labels for each observation.

Obs.	X_1	X_2	
1	1	4	●
2	1	3	●
3	0	4	●
4	5	1	●
5	6	2	●
6	4	0	●



Ex.3) K-mean clustering K=2 (c)

(c) Compute the centroid for each cluster.

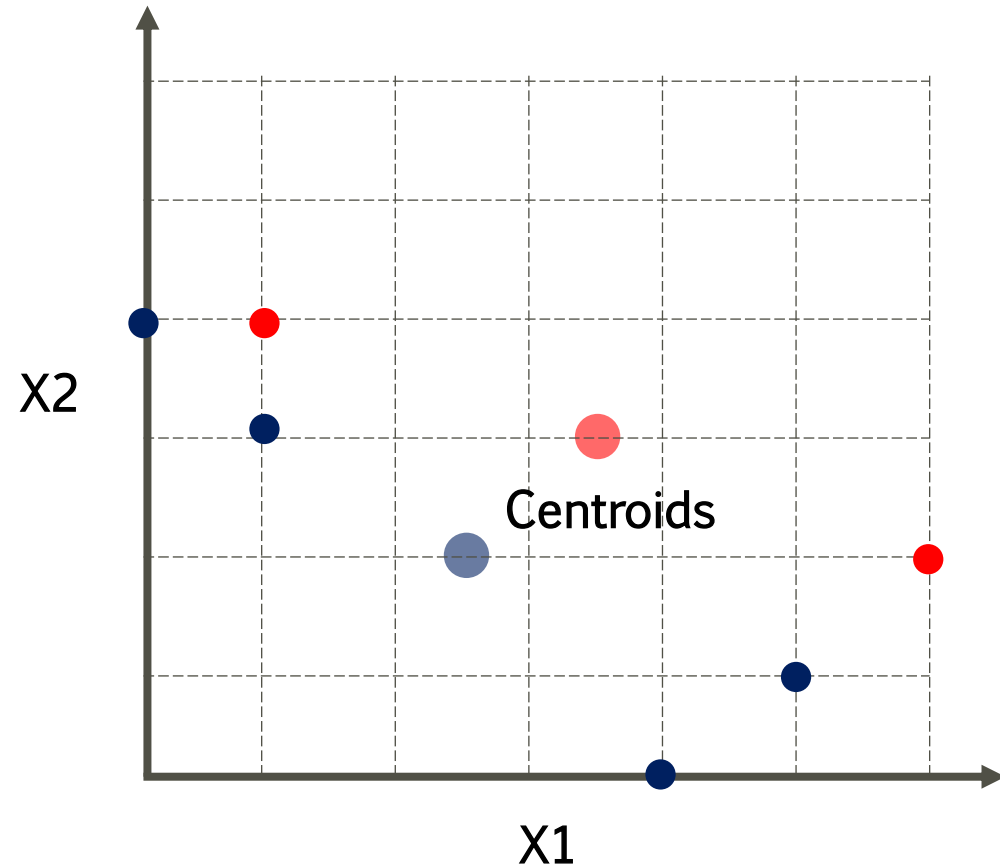
$$\mu_k = \left(\frac{1}{n_k} \sum_{i \in C_k} x_{i1}, \frac{1}{n_k} \sum_{i \in C_k} x_{i2}, \dots, \frac{1}{n_k} \sum_{i \in C_k} x_{ip} \right)$$

$$\mu_B = \frac{1}{4} (1 + 0 + 5 + 4 | 3 + 4 + 1 + 0)$$

$$\mu_B = \left(\frac{5}{2} \middle| 2 \right)$$

$$\mu_R = \frac{1}{2} (1 + 6 | 4 + 2)$$

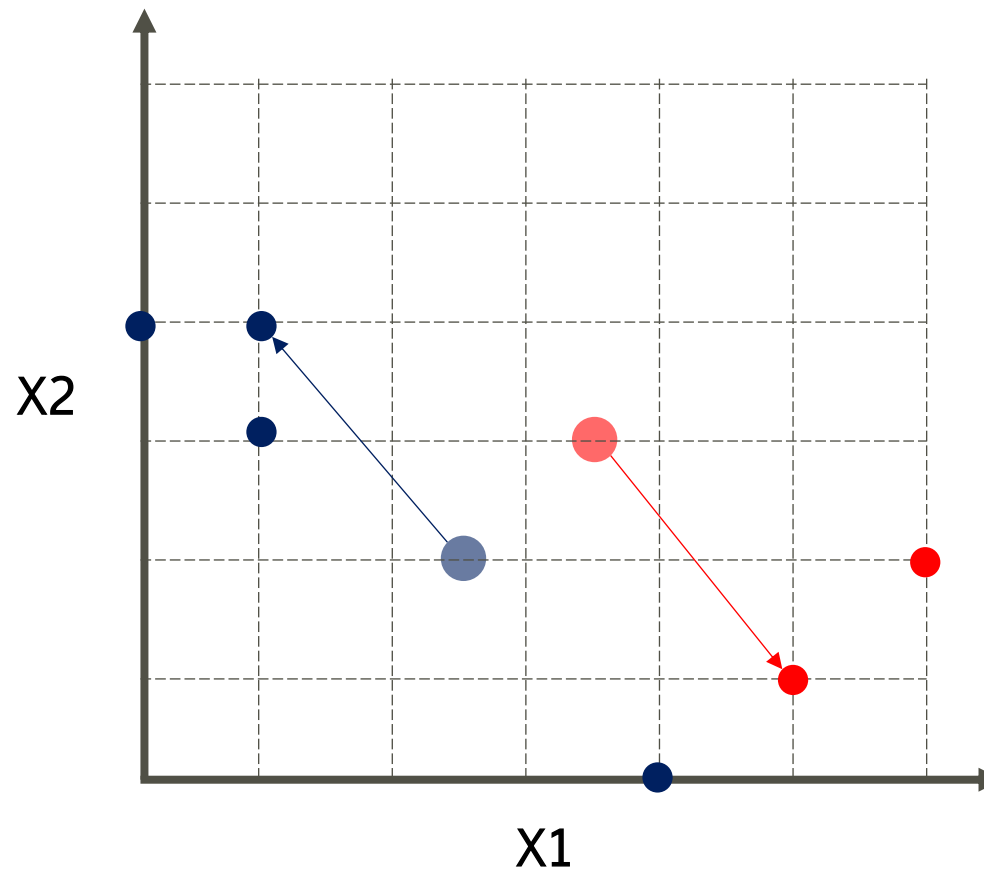
$$\mu_R = \left(\frac{7}{2} \middle| 3 \right)$$



Ex.3) K-mean clustering K=2 (d)

- (d) Assign each observation to the centroid to which it is closest, in terms of Euclidean distance. Report the cluster labels for each observation.

Obs.	X_1	X_2	1st	2nd
1	1	4	●	●
2	1	3	●	●
3	0	4	●	●
4	5	1	●	●
5	6	2	●	●
6	4	0	●	●



Ex.3) K-mean clustering K=2 (e)

(e) Repeat (c) and (d) until the answers obtained stop changing.

2nd iteration

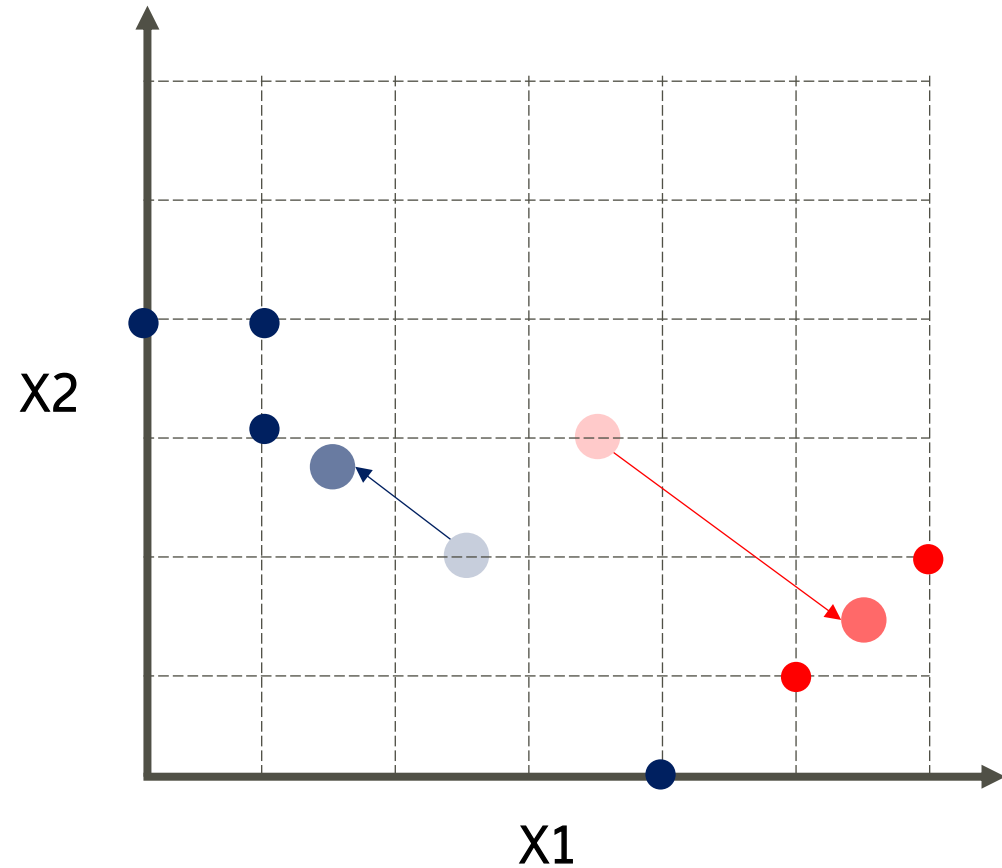
$$\mu_B = \frac{1}{4}(1 + 1 + 0 + 4 | 4 + 3 + 4 + 0)$$

$$\mu_B = \left(\frac{6}{4} \middle| \frac{11}{4} \right)$$



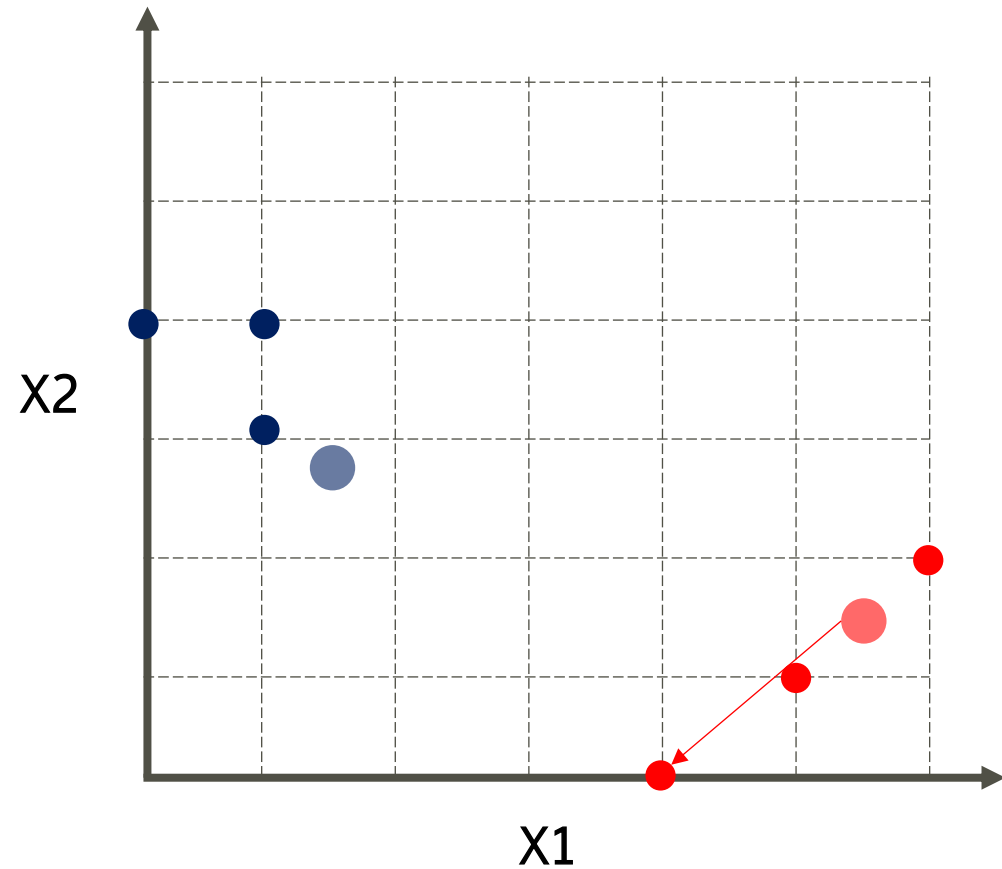
$$\mu_R = \frac{1}{2}(5 + 6 | 1 + 2)$$

$$\mu_R = \left(\frac{11}{2} \middle| \frac{3}{2} \right)$$



Ex.3) K-mean clustering K=2 (e)

Obs.	X_1	X_2	2nd	3rd
1	1	4	●	●
2	1	3	●	●
3	0	4	●	●
4	5	1	●	●
5	6	2	●	●
6	4	0	●	●



Ex.3) K-mean clustering K=2 (e)

3rd iteration

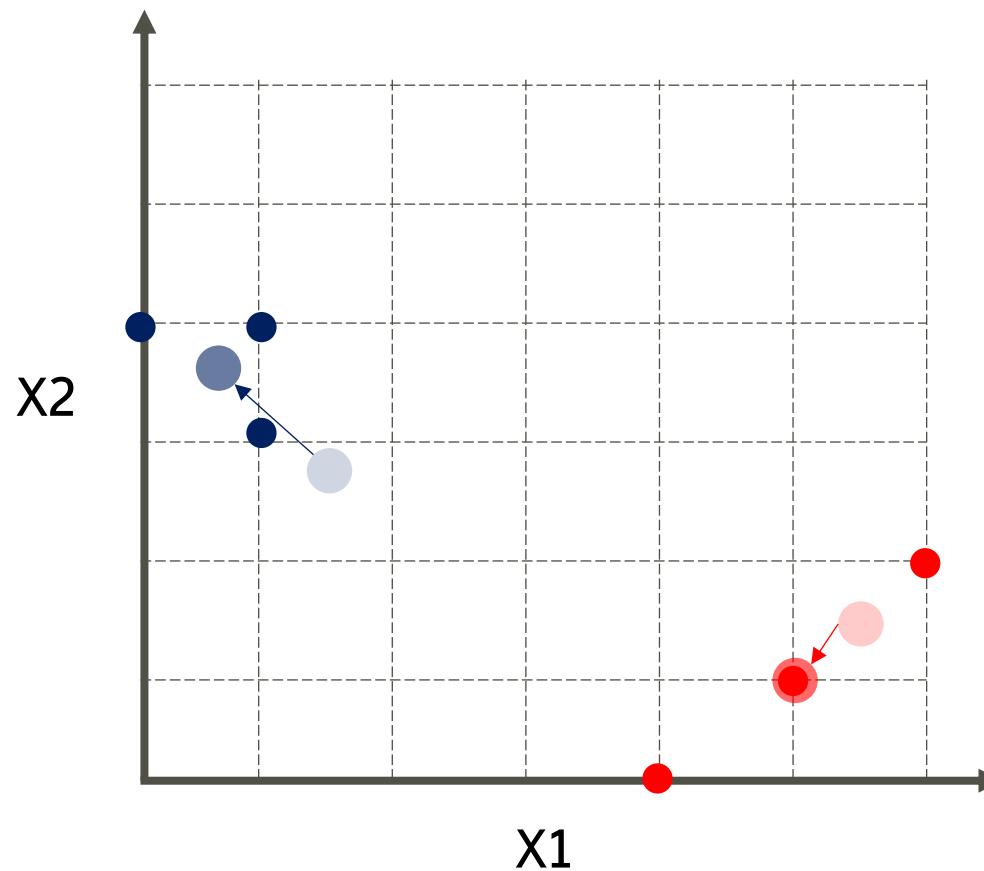
$$\mu_B = \frac{1}{3}(1 + 1 + 0 | 4 + 3 + 4)$$

$$\mu_B = \left(\frac{2}{3} \middle| \frac{11}{3} \right)$$



$$\mu_R = \frac{1}{3}(5 + 6 + 4 | 1 + 2 + 0)$$

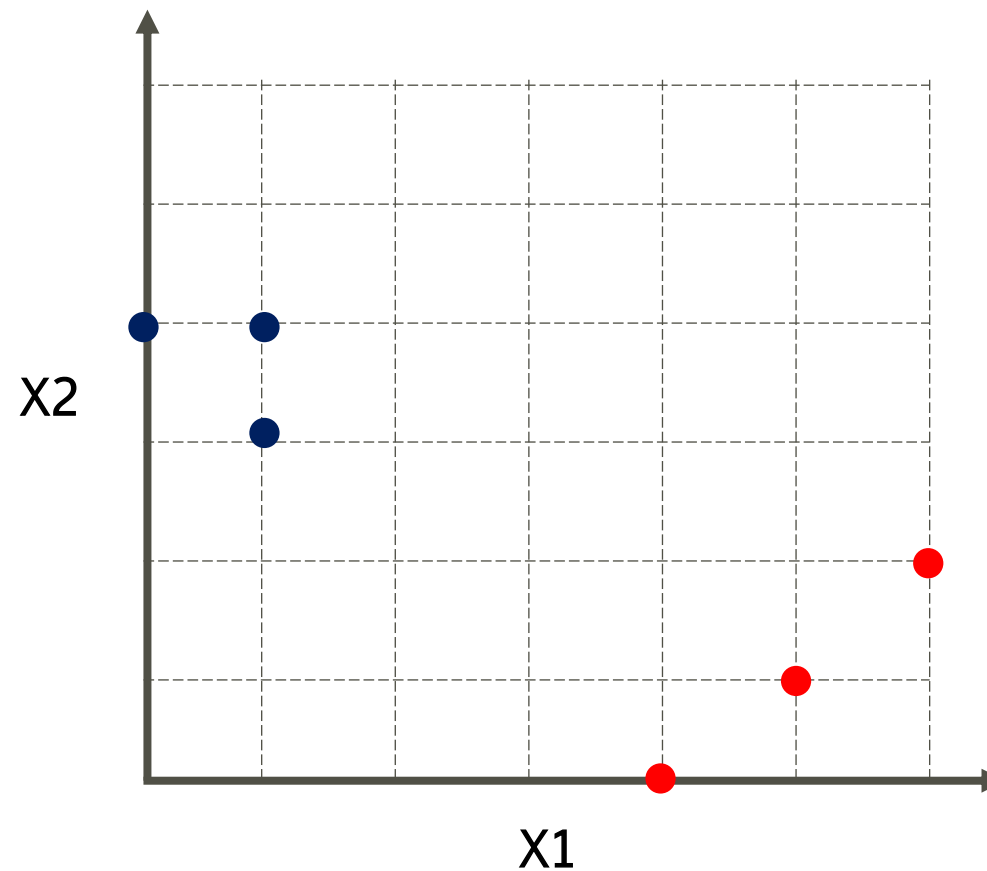
$$\mu_R = (5 | 1)$$



Ex.3) K-mean clustering K=2 (f)

(f) In your plot from (a), color the observations according to the cluster labels obtained.

Obs.	X_1	X_2	3rd
1	1	4	●
2	1	3	●
3	0	4	●
4	5	1	●
5	6	2	●
6	4	0	●





Applied

Exercise: 10

Ex.10) PCA and K-means clustering (a)

10. In this problem, you will generate simulated data, and then perform PCA and *K*-means clustering on the data.

- (a) Generate a simulated data set with 20 observations in each of three classes (i.e. 60 observations total), and 50 variables.

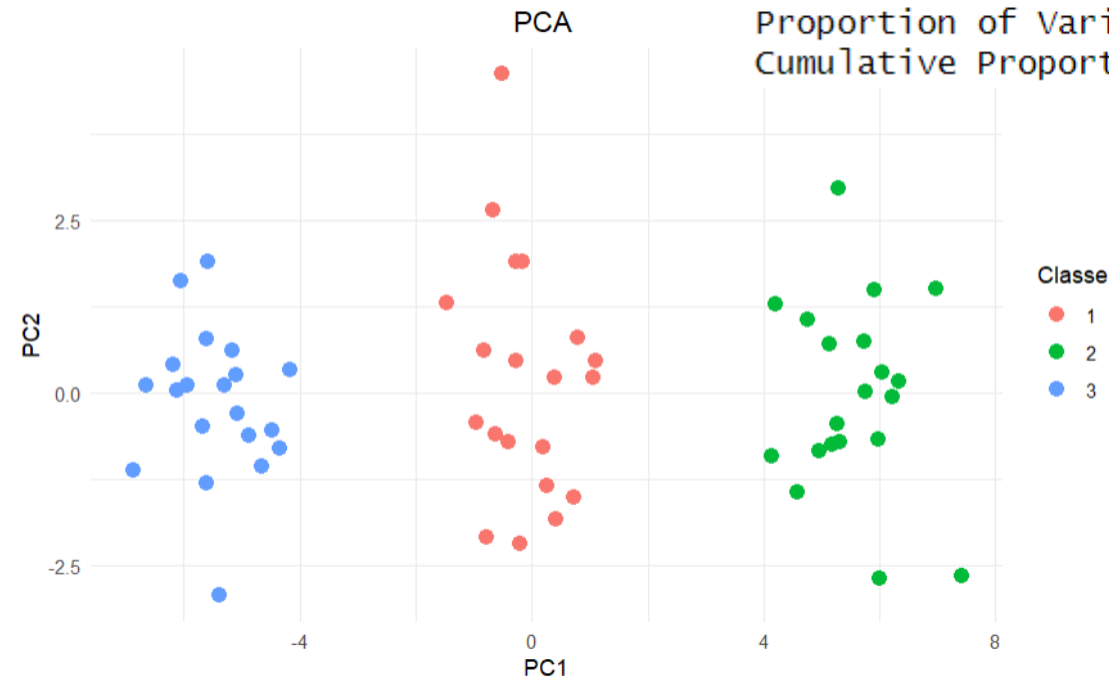
```
# Dados
set.seed(2367070)
n <- 20      # observations per class
p <- 50      # variables per observation
K <- 3       # number of classes

classe1 <- matrix(rnorm(n * p, mean = 0, sd = 1), nrow = n, ncol = p)
classe2 <- matrix(rnorm(n * p, mean = 1, sd = 1), nrow = n, ncol = p)
classe3 <- matrix(rnorm(n * p, mean = -1, sd = 1), nrow = n, ncol = p)
dados <- rbind(classe1, classe2, classe3)
classes <- factor(rep(1:3, each = n))
```

Ex.10) PCA and K-means clustering (b)

(b) Perform PCA on the 60 observations and plot the first two principal component score vectors. Use a different color to indicate the observations in each of the three classes. If the three classes appear separated in this plot, then continue on to part (c). If not, then return to part (a) and modify the simulation so that there is greater separation between the three classes. Do not continue to part (c) until the three classes show at least some separation in the first two principal component score vectors.

- PC1 – separates the classes given the mean shift
- PC2 – captures variability within the classes



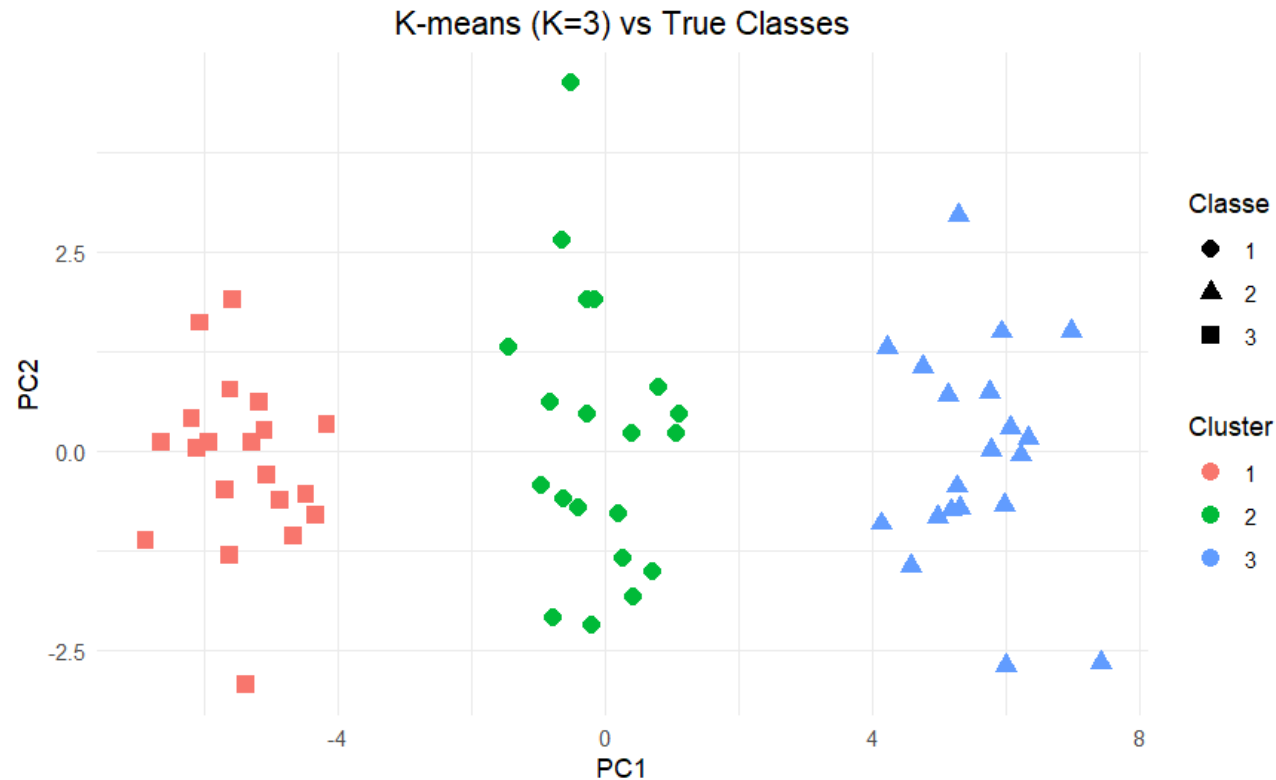
Importance of components:

	PC1	PC2
Standard deviation	4.5918	1.39798
Proportion of Variance	0.4217	0.03909
Cumulative Proportion	0.4217	0.46078

Ex.10) PCA and K-means clustering (c)

- (c) Perform K -means clustering of the observations with $K = 3$. How well do the clusters that you obtained in K -means clustering compare to the true class labels?

- 100% accuracy – no misclassification since there are 3 separable classes

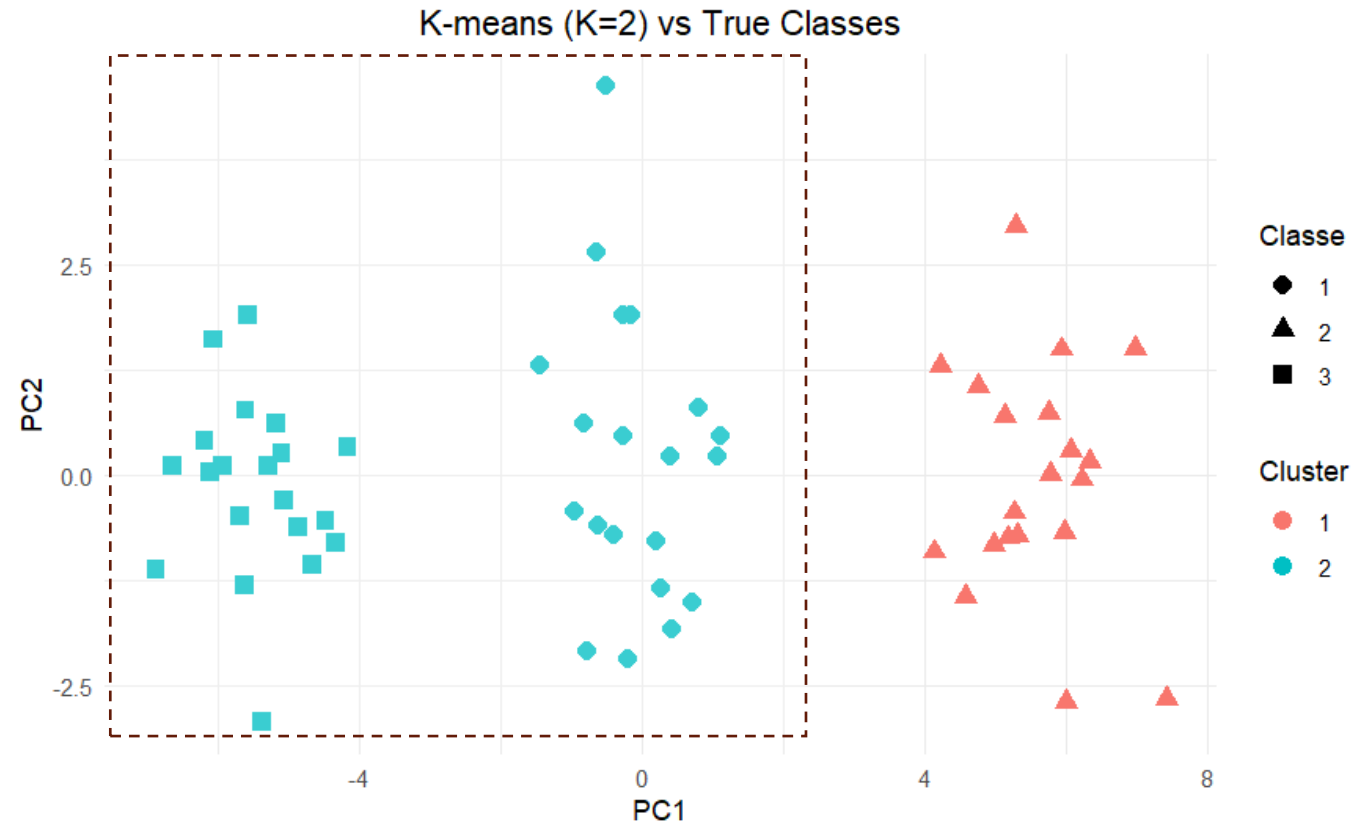


Predicted			
Real \	1	2	3
1	0	20	0
2	0	0	20
3	20	0	0

Ex.10) PCA and K-means clustering (d)

(d) Perform K -means clustering with $K = 2$. Describe your results.

- Two classes classified in the same cluster – wrong assumption on number of classes

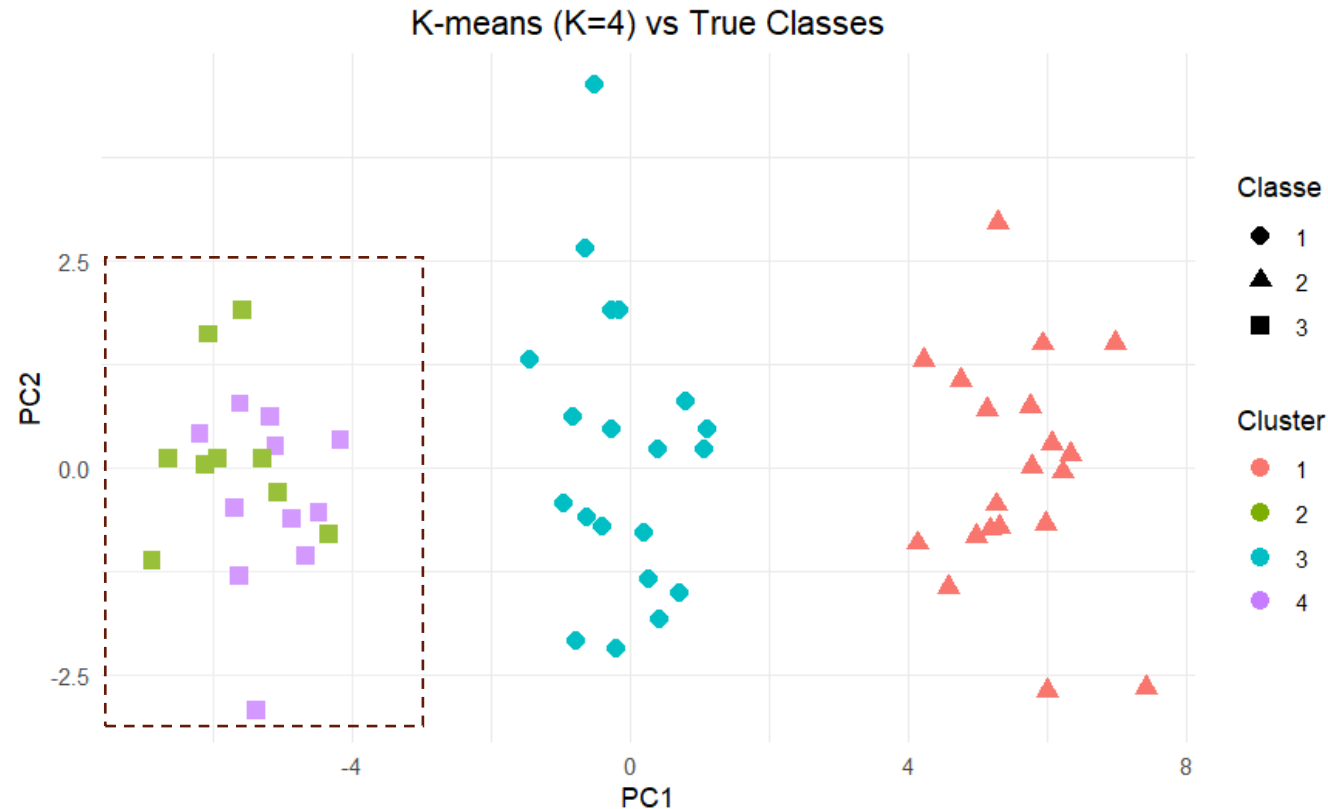


Predicted		
Real \	1	2
1	0	20
2	20	0
3	0	20

Ex.10) PCA and K-means clustering (e)

(e) Now perform K -means clustering with $K = 4$, and describe your results.

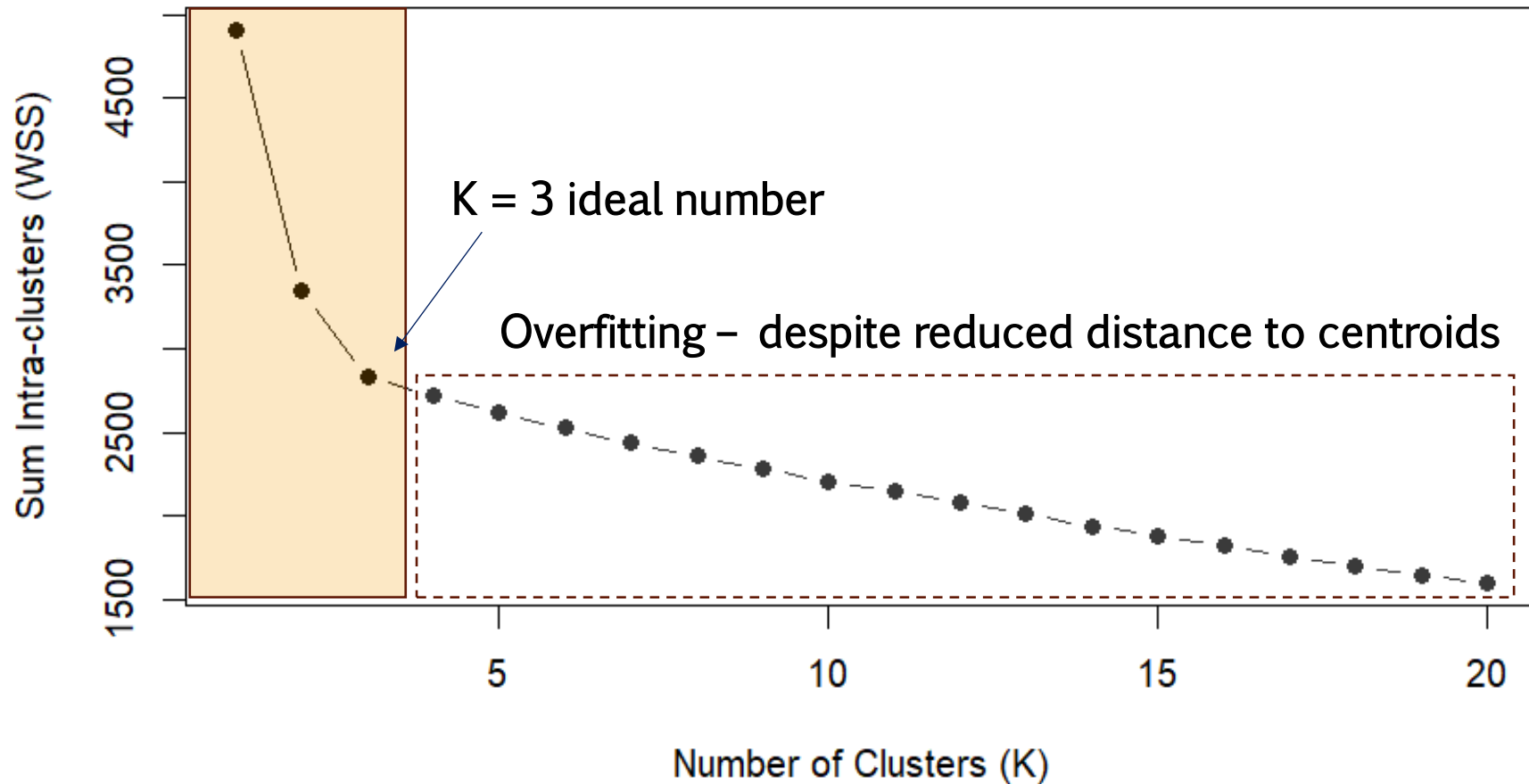
- One class divided in 2 clusters – importance of having a sense of number of clusters



Real	Predicted			
	1	2	3	4
1	0	0	20	0
2	20	0	0	0
3	0	9	0	11

Ex.10) PCA and K-means clustering (e)

Elbow chart



Ex.10) PCA and K-means clustering (all)

- In unsupervised learning, there is no previous knowledge on the real number of clusters (K)
- Results depends on the numbers of clusters: elbow charts are a useful tool to set the “ideal” number of clusters, since it avoids overfitting in more flexible models
- Clusters will normally be impacted by the data, so it is recommended to use standardized data